

React Cheat Sheet v16

☐ Component☐ Children☐ Context☐ react-dom☐ test-utils☐ misc

render

react

[docs](#)

```
render() {  
  return <div />;  
}
```



constructor

react

[docs](#)

```
constructor(props) {  
  super(props);  
  this.state = {  
    list: props.initialList  
  };  
}
```

// where props aren't used in constructor

```
constructor() {  
  super();  
  this.state = {  
    list: []  
  };  
}
```

componentWillMount

react

[docs](#)

```
componentWillMount() {  
  // invoked once.
```

```
// fires before initial 'render'  
}
```

componentDidMount

react

[docs](#)

```
componentDidMount() {  
  // good for AJAX: fetch, ajax, or subscriptions.  
  
  // invoked once (client-side only).  
  // fires before initial 'render'  
}
```



componentWillReceiveProps

react

[docs](#)

```
componentWillReceiveProps(nextProps) {  
  // invoked every time component receives new props.  
  // does not fire before initial 'render'  
}
```

shouldComponentUpdate

react

[docs](#)

```
shouldComponentUpdate(nextProps, nextState) {  
  // invoked before every update (new props or state).  
  // does not fire before initial 'render'.  
}
```

componentWillUpdate

react

[docs](#)

```
componentWillUpdate(nextProps, nextState) {  
  // invoked immediately before update (new props or state).  
  // does not fire before initial 'render'.  
}
```

```
// (see componentWillReceiveProps if you need to call setState)
}
```

✖ **this.setState**

componentDidUpdate

react

[docs](#)

```
componentDidUpdate(prevProps, prevState) {
  // invoked immediately after DOM updates
  // does not fire after initial 'render'
}
```



componentWillUnmount

react

[docs](#)

```
componentWillUnmount() {
  // invoked immediately before a component is unmounted.
}
```

setState (function)

react

[docs](#)

```
// good for state transitions

this.setState((prevState, props) => {
  return {count: prevState.count + props.step};
});
```

setState (object)

react

[docs](#)

```
// good for static values

this.setState({mykey: 'my new value'});
```

setState (optional callback)

react

[docs](#)

```
// fires after setState
// prefer componentDidUpdate

this.setState(
  (prevState, props) => ({ count: prevState.count + props.step }),
  () => console.log(this.state.count)
);
```

forceUpdate

react



```
// forces a re-render; AVOID if possible

this.forceUpdate();
```

displayName

react

[docs](#)

```
displayName: "MyComponent"
```

defaultProps

react

[docs](#)

```
class Greeting extends React.Component {
  render() {
    return <h1>Hi {this.props.name}</h1>
  }
}

CustomButton.defaultProps = {
  name: 'guest'
};
```

Children.map

react

[docs](#)

```
React.Children.map(this.props.children, (child, i) => {  
  return child;  
})
```

Children.forEach

react

[docs](#)

```
React.Children.forEach(this.props.children, (child, i) => {  
  console.log(child + ' at index: ' + i);  
})
```



Children.count

react

[docs](#)

```
React.Children.count(this.props.children);
```

Children.only

react

[docs](#)

```
React.Children.only(this.props.children);
```

Children.toArray

react

[docs](#)

```
React.Children.toArray(this.props.children)
```

Context (example)

react

[docs](#)

```
// requires 'prop-types' library

import { string } from "prop-types";

class Cowboy extends React.Component {
  childContextTypes: {
    salutation: string
  }

  getChildContext() {
    return { salutation: "Howdy" };
  }

  render() {
    return React.Children.only(this.props.children);
  }
}

const Greeting = (props, context) =>
  <div>{context.salutation} {props.name}</div>

Greeting.contextTypes = {
  salutation: PropTypes.string
}

// <Greeting name="Michael" />
// => Michael.

// <Cowboy><Greeting name="Michael" /></Cowboy>
// => Howdy Michael.
```



contextTypes

react

[docs](#)

```
// add to the context-aware component
// requires 'prop-types' library

contextTypes: {
  color: PropTypes.string
},
```

childContextTypes

react

[docs](#)

```
// add to the context provider
// requires 'prop-types' library
```

```
childContextTypes: {  
  color: PropTypes.string  
},
```

getChildContext

react

[docs](#)

```
// add to the context provider  
  
getChildContext() {  
  return {color: "purple"};  
}
```



render

react-dom

[docs](#)

```
import { render } from "react-dom";  
  
render(  
  <MyComponent />,  
  document.getElementById("component-root"),  
  () => console.log("MyComponent mounted.")  
);
```

hydrate

react-dom

[docs](#)

```
import { hydrate } from "react-dom";  
  
hydrate(  
  <MyComponent />,  
  document.getElementById("component-root"),  
  () => console.log("MyComponent hydrated.")  
);
```

unmountComponentAtNode

react-dom

[docs](#)

```
import { unmountComponentAtNode } from "react-dom";

unmountComponentAtNode(document.getElementById('MyComponent'))
```

findDOMNode

react-dom

[docs](#)

```
import { findDOMNode } from "react-dom";

findDOMNode(componentRef);
```



createPortal

react-dom

[docs](#)

```
import { createPortal } from "react-dom";

class MyPortalComponent extends React.Component {
  render() {
    return createPortal(
      this.props.children,
      document.getElementById("portal-element"),
    );
  }
}
```

renderToString

react-dom/server

[docs](#)

```
import { renderToString } from "react-dom/server";
ReactDOMServer.renderToString(<MyComponent />);
```

renderToStaticMarkup

react-dom/server

[docs](#)

```
import { renderToStaticMarkup } from "react-dom/server";
```



```
renderToStaticMarkup(<MyComponent />);
```

renderToNodeStream

react-dom/server

[docs](#)

```
import { renderToNodeStream } from "react-dom/server";
renderToNodeStream(<MyComponent />);
```

renderToStaticNodeStream

react-dom/server



```
import { renderToStaticNodeStream } from "react-dom/server";
renderToStaticNodeStream(<MyComponent />);
```

Simulate (basic)

react-dom/test-utils

[example](#) [docs](#)

```
var subject = TestUtils.renderIntoDocument(
  <div onClick={handleClick} />
);

TestUtils.Simulate.click(subject);
```

Simulate (with data)

react-dom/test-utils

[example](#) [docs](#)

```
function handleChange (event) {
  console.log('A change was simulated with key: ' + event.key);
}

var subject = TestUtils.renderIntoDocument(
  <input type="text" onChange={handleChange} />
);

TestUtils.Simulate.change(subject, { key: "Enter" });
```

renderIntoDocument

react-dom/test-utils

[example](#) [docs](#)

```
var componentTree = TestUtils.renderIntoDocument(<div><span /></div>);  
console.log('You mounted a component tree with a ' + componentTree.tag
```

mockComponent

react-dom/test-utils

[docs](#)

```
// no example
```



isElement

react-dom/test-utils

[example](#) [docs](#)

```
expect(TestUtils.isElement(<div />)).toBe(true);
```

isElementOfType

react-dom/test-utils

[example](#) [docs](#)

```
var MyComponent = React.createClass({  
  render () {  
    return <div />;  
  }  
});  
  
expect(  
  TestUtils.isElementOfType(<MyComponent />, MyComponent)  
).toBe(true);
```

isDOMComponent

react-dom/test-utils

[example](#) [docs](#)

```
var subject = TestUtils.renderIntoDocument(<div />);

expect(
  TestUtils.isDOMComponent(subject)
).toBe(true);
```

isCompositeComponent

react-dom/test-utils

[example](#) [docs](#)

```
var subject = TestUtils.renderIntoDocument(
  <CompositeComponent />
);

expect(
  TestUtils.isCompositeComponent(subject)
).toBe(true);
```



isCompositeComponentWithType

react-dom/test-utils

[example](#) [docs](#)

```
var CompositeComponent = React.createClass({
  render () {
    return <div />;
  }
});

var subject = TestUtils.renderIntoDocument(
  <CompositeComponent />
);

expect(
  TestUtils.isCompositeComponentWithType(
    subject,
    CompositeComponent
  )
).toBe(true);
```

findAllInRenderedTree

react-dom/test-utils

[example](#) [docs](#)

```
var CompositeComponent = React.createClass({
  render () {
```

```
        return <div><div /></div>;
    }
  });

  var componentTree = TestUtils.renderIntoDocument(
    <CompositeComponent />
  );

  var allDivs = TestUtils.findAllInRenderedTree(
    componentTree,
    (c) => c.tagName === 'DIV'
  )

  expect(allDivs).toBeAn('array');
  expect(allDivs.length).toBe(2);
```



scopyRenderedDOMComponentsWithClass

react-dom/test-utils

[example](#) [docs](#)

```
var CompositeComponent = React.createClass({
  render () {
    return (
      <div className="target">
        <div className="not-target">
          <div className="target" />
        </div>
      </div>
    );
  }
});

var componentTree = TestUtils.renderIntoDocument(
  <CompositeComponent />
);

var allDOMComponentsWithMatchingClass = TestUtils.scopyRenderedDOMComp
  componentTree,
  'target'
);

expect(allDOMComponentsWithMatchingClass).toBeAn('array');
expect(allDOMComponentsWithMatchingClass.length).toBe(2);
```

findRenderedDOMComponentWithClass

react-dom/test-utils

[example](#) [docs](#)

```

var MyCompositeComponent = React.createClass({
  render () {
    return <MyNestedComponent />;
  }
});

var MyNestedComponent = React.createClass({
  render () {
    return <div className="nested"/>;
  }
});

var componentTree = TestUtils.renderIntoDocument(<MyCompositeComponent

var singleComponentWithMatchedClass = TestUtils.findRenderedDOMCompone
  componentTree,
  'nested'
);

expect(singleComponentWithMatchedClass).toBeAn('object');
expect(singleComponentWithMatchedClass).toNotBeAn('array');
expect(singleComponentWithMatchedClass.className).toBe('nested');

```



scryRenderedDOMComponentsWithTag

react-dom/test-utils

[example](#) [docs](#)

```

var CompositeComponent = React.createClass({
  render () {
    return <div><div /></div>;
  }
});

var componentTree = TestUtils.renderIntoDocument(
  <CompositeComponent />
);

var allDivs = TestUtils.scryRenderedDOMComponentsWithTag(
  componentTree,
  'DIV'
);

expect(allDivs).toBeAn('array');
expect(allDivs.length).toBe(2);

```

findRenderedDOMComponentWithTag

react-dom/test-utils

[example](#) [docs](#)

```

var MyCompositeComponent = React.createClass({
  render () {
    return <MyNestedComponent />;
  }
});

var MyNestedComponent = React.createClass({
  render () {
    return <div />;
  }
});

var componentTree = TestUtils.renderIntoDocument(<MyCompositeComponent />);

var onlyDiv = TestUtils.findRenderedDOMComponentWithTag(
  componentTree,
  'div'
);

expect(onlyDiv).toBeAn('object');
expect(onlyDiv).toBeAn('array');
expect(onlyDiv.tagName).toBe('DIV');

```



scryRenderedComponentsWithType

react-dom/test-utils

[example](#) [docs](#)

```

var MyCompositeComponent = React.createClass({
  render () {
    return (
      <div>
        <Target />
        <br />
        <Target />
      </div>
    )
  }
});

var Target = React.createClass({
  render () {
    return <div />;
  }
});

var componentTree = TestUtils.renderIntoDocument(
  <MyCompositeComponent />
);

var allTargetComponents = TestUtils.scryRenderedComponentsWithType(
  componentTree,

```

```
    Target  
  );
```

```
expect(allTargetComponents).toBeAn('array');  
expect(allTargetComponents.length).toBe(2);
```

findRenderedComponentWithType

react-dom/test-utils

[example](#) [docs](#)

```
var MyCompositeComponent = React.createClass({  
  render () { return <TargetComponent /> }  
});  
  
var TargetComponent = React.createClass({  
  render () { return <div /> }  
});  
  
var componentTree = TestUtils.renderIntoDocument(  
  <MyCompositeComponent />  
);  
  
var onlyTargetComponent = TestUtils.findRenderedComponentWithType(  
  componentTree,  
  TargetComponent  
);  
  
expect(onlyTargetComponent).toBeAn('object');  
expect(onlyTargetComponent).toNotBeAn('array');  
expect(TestUtils.isCompositeComponentWithType(  
  onlyTargetComponent,  
  TargetComponent  
)).toBe(true);
```



Shallow rendering (basics)

react-dom/test-utils

[example](#) [docs](#)

```
// 1. create a renderer  
var renderer = TestUtils.createRenderer();  
  
// 2. render component into renderer  
renderer.render(<MyComponent />);  
  
// 3. capture renderer output  
var subject = renderer.getRenderOutput();
```

```
// 4. make assertions
expect(subject.type).toBe('div');
```

Shallow rendering (type example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

renderer.render(<MyComponent />);

var subject = renderer.getRenderOutput();

expect(subject.type).toBe('div'); // => true
```



Shallow rendering (props example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

renderer.render(<MyComponent className="my-component" />);

var subject = renderer.getRenderOutput();

expect(subject.props.className).toBe('my-component'); // => true
```

Shallow rendering (child-count example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

renderer.render(
  <MyList items={[1, 2, 3]} />
);

var subject = renderer.getRenderOutput();

var childCount = React.Children.count(subject.props.children);

expect(childCount).toBe(3); // => true
```

Shallow rendering (child-equality example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

renderer.render(
  <MyComponent>
    <div>Thing 1</div>
    <div>Thing 2</div>
  </MyComponent>
);

var subject = renderer.getRenderOutput();

expect(subject.props.children).toEqual([
  <div>Thing 1</div>,
  <div>Thing 2</div>
]); // => true
```



Shallow rendering (events example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

var spy = expect.createSpy();

renderer.render(<MyComponent onClick={spy} />);

var subject = renderer.getRenderOutput();

expect(spy.call.length).toEqual(1); // => true
```

Shallow rendering (state changes example)

react-dom/test-utils

[example](#) [docs](#)

```
var renderer = TestUtils.createRenderer();

renderer.render(<ClickCounter />);

// test initial rendering
var result = renderer.getRenderOutput();

expect(result.props.children).toEqual(0);
```

```
// test post-click rendering
result.props.onClick();

var clickedResult = renderer.getRenderOutput();

expect(clickedResult.props.children).toEqual(1);
```

Lifecycle methods

NAME	CALLED FOR	RECEIVES CONTEXT	SETSTATE() TRIGGERS
componentWillMount	initial render()	no	no
componentDidMount	initial render()	no	yes
componentWillReceiveProps	new props	yes	yes
shouldComponentUpdate	new props/state	yes	yes
componentWillUpdate	new props/state	yes	no
componentDidUpdate	new props/state	no	yes
componentWillUnmount	unmounting	no	no

ref (class component)

[docs](#)

```
class AutoFocusTextInput extends React.Component {
  componentDidMount() {
    this.textInput.focus();
  }

  render() {
    return (
      <CustomTextInput
        ref={(input) => { this.textInput = input; }} />
    );
  }
}
```

ref (functional component)

[docs](#)

```
function CustomTextInput(props) {
  let textInput = null;

  function handleClick() {
    textInput.focus();
  }

  return (
    <div>
      <input
        type="text"
        ref={(input) => { textInput = input; }} />
      <input
        type="button"
        value="Focus the text input"
        onClick={handleClick}
      />
    </div>
  );
}
```



functional component

react

[docs](#)

```
const Greeting = props => <div>Hello {props.name}</div>;
```

functional component (with context)

react

[docs](#)

```
import { string } from "prop-types";

const Greeting = (props, context) => (
  <div>{context.salutation} {props.name}</div>
);

Greeting.contextTypes = { salutation: string };
```



Learn all about **functional components** in a new course by Learn React.

